# Queue Development Planning Guide

## Purpose

This document serves as the master implementation plan for Queue.

The objective is to build a production-quality desktop application while minimizing scope creep and ensuring steady progress.

Development should follow a milestone-based approach.

Success is defined as shipping a usable product rather than implementing every possible feature.

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# Development Principles

1. Build core functionality first.
2. Ship early.
3. Avoid premature AI integration.
4. Prioritize reliability over feature count.
5. Keep platform-specific code isolated.
6. Every feature must solve a real user problem.

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# Phase 0 - Discovery and Technical Decisions

Goal:

Finalize architecture before writing major code.

Tasks:

- Choose frontend framework.
- Choose desktop framework.
- Choose storage solution.
- Research clipboard APIs.
- Research global hotkey APIs.
- Research mouse button capture.
- Research keyboard simulation.
- Define project structure.
- Define coding standards.

Deliverable:

Technical Architecture Document.

Acceptance Criteria:

- Architecture documented.
- Repository initialized.
- Development environment working.

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# Phase 1 - Project Skeleton

Goal:

Create the application foundation.

Tasks:

- Initialize repository.
- Configure linting.
- Configure formatting.
- Configure CI pipeline.
- Create application shell.
- Create navigation system.
- Create settings system.
- Create persistent storage layer.

Deliverable:

Launchable desktop application.

Acceptance Criteria:

- Application launches.
- Navigation works.
- Settings persist after restart.

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# Phase 2 - Queue Clipboard MVP

Goal:

Implement the core Queue concept.

User Story:

As a user,

I want to store multiple copied items,

so I can paste them sequentially later.

Tasks:

- Clipboard monitoring service.
- Queue state management.
- Queue UI.
- Sequential paste logic.
- Queue clearing.
- Queue persistence.

Acceptance Criteria:

- Multiple items can be added.
- Sequential paste works.
- Queue survives restart.
- Clear Queue works.

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# Phase 3 - Clipboard Slots

Goal:

Implement persistent clipboard slots.

User Story:

As a user,

I want dedicated clipboard locations,

so I can instantly access frequently used information.

Tasks:

- Slot storage system.
- Slot UI.
- Slot editing.
- Slot deletion.
- Slot persistence.
- Slot hotkeys.

Acceptance Criteria:

- Users can create slots.
- Users can rename slots.
- Users can paste from slots.
- Slots survive restart.

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# Phase 4 - Global Hotkeys

Goal:

Create customizable shortcuts.

User Story:

As a user,

I want custom shortcuts,

so Queue fits my workflow.

Tasks:

- Global shortcut service.
- Shortcut recorder.
- Conflict detection.
- Hotkey settings UI.
- Shortcut persistence.

Acceptance Criteria:

- Users can rebind shortcuts.
- Rebindings survive restart.
- Global shortcuts work outside Queue.

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# Phase 5 - Profiles

Goal:

Implement reusable information collections.

User Story:

As a user,

I want reusable profiles,

so I can quickly fill repetitive forms.

Tasks:

- Profile model.
- Profile editor.
- Profile manager.
- Profile loading.
- Queue population.

Acceptance Criteria:

- Profiles can be created.
- Profiles can be edited.
- Profiles can populate Queue.
- Profiles persist.

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# Phase 6 - History

Goal:

Implement searchable clipboard history.

Tasks:

- History database.
- Search indexing.
- Search UI.
- Pinning system.
- Favorites system.

Acceptance Criteria:

- Users can search history.
- Users can pin items.
- Users can restore items.

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# Phase 7 - Onboarding Experience

Goal:

Improve first-run experience.

Tasks:

- Welcome page.
- Feature overview.
- Shortcut setup.
- Preferences setup.
- Completion screen.

Acceptance Criteria:

- New users can complete setup.
- Preferences save correctly.
- Hotkeys can be assigned.

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# Phase 8 - Intelligence Layer

Goal:

Add productivity-enhancing intelligence.

User Story:

As a user,

I want Queue to recognize common data types,

so I can work faster.

Tasks:

- Email detection.
- URL detection.
- Phone detection.
- Date detection.
- Address detection.
- Classification UI.

Acceptance Criteria:

- Data types detected accurately.
- Suggestions are relevant.
- Classification never blocks normal use.

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# Phase 9 - Profile Suggestions

Goal:

Generate reusable profiles automatically.

Tasks:

- Pattern detection.
- Frequency tracking.
- Suggestion engine.
- Profile generation flow.

Acceptance Criteria:

- Repeated patterns detected.
- Suggestions generated correctly.
- Users can accept or reject suggestions.

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# Phase 10 - Release Preparation

Goal:

Prepare for public release.

Tasks:

- Error handling.
- Logging.
- Crash reporting.
- Installer creation.
- Documentation.
- User guide.
- Privacy policy.
- Website.

Acceptance Criteria:

- Stable release build.
- Installation process tested.
- Documentation complete.

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# Stretch Goals

Not required for Version 1.

Possible future features:

- Cloud sync
- Mobile companion
- Team profiles
- AI-powered autofill
- Workflow recording
- Command palette
- Local AI models
- Plugin system

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# Definition of Done

Queue Version 1 is complete when:

- Queue Clipboard works reliably.
- Clipboard Slots work reliably.
- Profiles work reliably.
- Hotkeys are customizable.
- History is searchable.
- Settings persist.
- Application feels polished.
- Application can be used daily.

At this point the project transitions from a portfolio project to a real product.